

DAILY RHYTHMS OF BODY TEMPERATURE IN COEXISTING RODENTS OF THE GENUS *ACOMYS* ACCLIMATED TO LONG-PHOTOPERIOD: EFFECTS OF ETHANOL AND MELATONIN

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Introduction

The daily rhythm of body temperature in endothermic animals is the product of two different rhythms, those of heat production and heat dissipation (2). While nocturnal species exhibit a daily rhythm of body temperature with maximal body temperature (Tb) values during the scotophase, diurnal ones increase their Tb during day time with maximal values at mid-day. Such a daily rhythm ensures that during their activity time, body temperature (Tb) is high while, during the resting phase, it is kept at relatively lower values, due to a decrease in heat production. Using such a "strategy", reduces daily energy expenditure (8).

An important behavioural adaptation in small mammals, such as rodents inhabiting arid environments, is their nocturnal activity. An exception to this activity pattern is found in the golden spiny mouse (*Acomys russatus*). This species, which is distributed throughout the extremely dry and hot regions of the Israeli Rift Valley, is diurnal (6). The diurnal activity of this species in these places is due to its exclusion from nocturnal activity by the common spiny mouse *A. cahirinus*. The removal of the latter from the common habitat results in nocturnal activity of the former (12). However, under controlled laboratory conditions, indirect introduction through an air inflow of diluted chemical signals released by *A. cahirinus* to nocturnal *A. russatus* resulted in a phase shift of 2h (3). Direct contact with chemical signals (urine and faeces) released by *A. cahirinus* advanced onset of activity of *A. russatus* by about 6h (4).

The effects of daily injections of melatonin and 7% ethanol in saline (used as a vehicle) on daily rhythms of Tb were studied previously in *A. russatus*, acclimated to a long photoperiod (16L:8D) at an ambient temperature (Ta) of 28°C (5). The results of that study showed, that under 16L:8D *A. russatus* exhibited circadian rhythmicity of Tb of a nocturnal species with maximal values recorded at 24:00h and minimal ones recorded at 12:00h. Daily s.c. injections (at 16:00h) for three days of 0.2ml of 7% ethanol in saline (vehicle) caused hyperthermia with maximal Tb values recorded at 12:00h. Similar injections of 50µg melatonin in 0.2ml vehicle for eight days abolished

the daily Tb rhythm (Fig. 2). Daily injections of 0.2ml saline used as control had no effect on the daily rhythm of Tb in long photoperiod acclimated *A. russatus*.

The aim of the present research was to study the responses of long photoperiod (16L:8D) acclimated *A. cahirinus* (the nocturnal species) to ethanol and melatonin, injection and compare them with those of *A. russatus*.

Material and Methods

Seven *A. cahirinus* (males, mean body mass 41.5 ± 3.2 g) from our breeding colony (University of Haifa at Oranim) were acclimated for 3-5 weeks to a long photoperiod 16L:8D (lights on between 06:00-22:00h) at $T_a = 28 \pm 1^\circ\text{C}$, (Forma Scientific). During the photophase cool white fluorescent illumination and red dim light were used while during scotophase only the red dim light was used. Each individual was kept in a separate cage and was supplied with rat pellets and fresh carrots as a water source ad lib. Sawdust was used for bedding. Body (rectal) temperature Tb was measured by using a copper-constantan thermocouple, and monitored on a TH-65 Wescor thermometer. The thermocouple was inserted to a depth of 3cm for no more than 30sec. Preliminary tests had shown that a daily injection of 0.2ml saline given at 16:00h did not cause any change in the daily rhythm of Tb.

After acclimation, the daily rhythm of Tb in such mice was recorded by measuring Tb at intervals of 6h for 72h. The same mice were injected daily, for three days s.c., with a vehicle (0.2ml of 7% ethanol in saline) at 16:00h, 6h before lights went off. Tb in such mice was then recorded at 6h intervals over 48h. Finally, the mice were injected daily, at 16:00h for eight days, with 50µg melatonin in 0.2ml vehicle and Tb was then recorded every 6h for 72h.

All results are given as mean $\pm$ SD. Student's t-test was used for statistical analysis.

Results

Common spiny mice *A. cahirinus* acclimated to a long photophase showed a daily rhythm of Tb. Maximal Tb values ($37.8 \pm 0.3^\circ\text{C}$) were recorded at 24:00h, while minimal Tb values (36.4 ± 0.4 and $36.5 \pm 0.3^\circ\text{C}$) were recorded at 12:00 and 18:00h respectively. Daily injections of 7% ethanol for three days caused a transient hypothermia in the first 24h after the treatment. However, Tb values returned to control values within 48h after the treatment. The melatonin treatment had no effect on the phase of the Tb daily rhythm of *A. cahirinus*, but increased the amplitude by lowering the minimal Tb value at 18:00h. As a result the difference between the maximal Tb values obtained at 24:00 and 06:00h and the minimal value recorded at 18:00h, in the melatonin injected mice (2.4°C) was higher than that in the control (1.4) (Fig 1). The daily rhythms of Tb of the golden spiny mouse *A. russatus* under the same treatments are shown in Fig. 2.

Discussion

A daily pattern of Tb variations is common among endothermic vertebrates. High Tbs are maintained during activity, while low values are recorded during periods of rest (9). Interestingly,